

Repository: - <https://github.com/ust-sdet-training/SDET/tree/aravind-25-06-2026>

Objective

To Perform API testing on <https://jsonplaceholder.typicode.com/>

Tests Performed

Test 1: Create a new Post using the Post method

(<https://jsonplaceholder.typicode.com/posts/>)

- Prepare the required payload and headers
- Make the post method call and verify the response using the schema and assertions
- 201 response

Test 2: Fetch a post using the get method providing the id

(<https://jsonplaceholder.typicode.com/posts/{id}>)

- Prepare the required headers
- Make the get method call and verify the response using the schema and assertions
- 200 response

Test 3: Delete a Post using the Delete method

(<https://jsonplaceholder.typicode.com/posts/{id}>)

- Prepare the required headers
- Make the delete method call and verify the response
- 200 response

Test 4: Perform a Fetch operation on an invalid id using the get method

(<https://jsonplaceholder.typicode.com/posts/{id}>)

- Prepare the required headers
- Make the get method call and verify the response code
- 404 response

Test 5: Update an existing post using the Put method

(<https://jsonplaceholder.typicode.com/posts/{id}>)

- Prepare the required payload and headers
- Make the put method call and verify the response using the schema and assertions
- 200 response

Test 6: Update an existing post using the Patch method

(<https://jsonplaceholder.typicode.com/posts/{id}>)

- Prepare the required payload and headers
- Make the patch method call and verify the response using the schema and assertions
- 200 response

Sample Data (Payload)

```
{  
  "userId": 101,  
  "id": 101,  
  "title": "hello",  
  "body": "hello123"  
}
```

Schema file for assertion

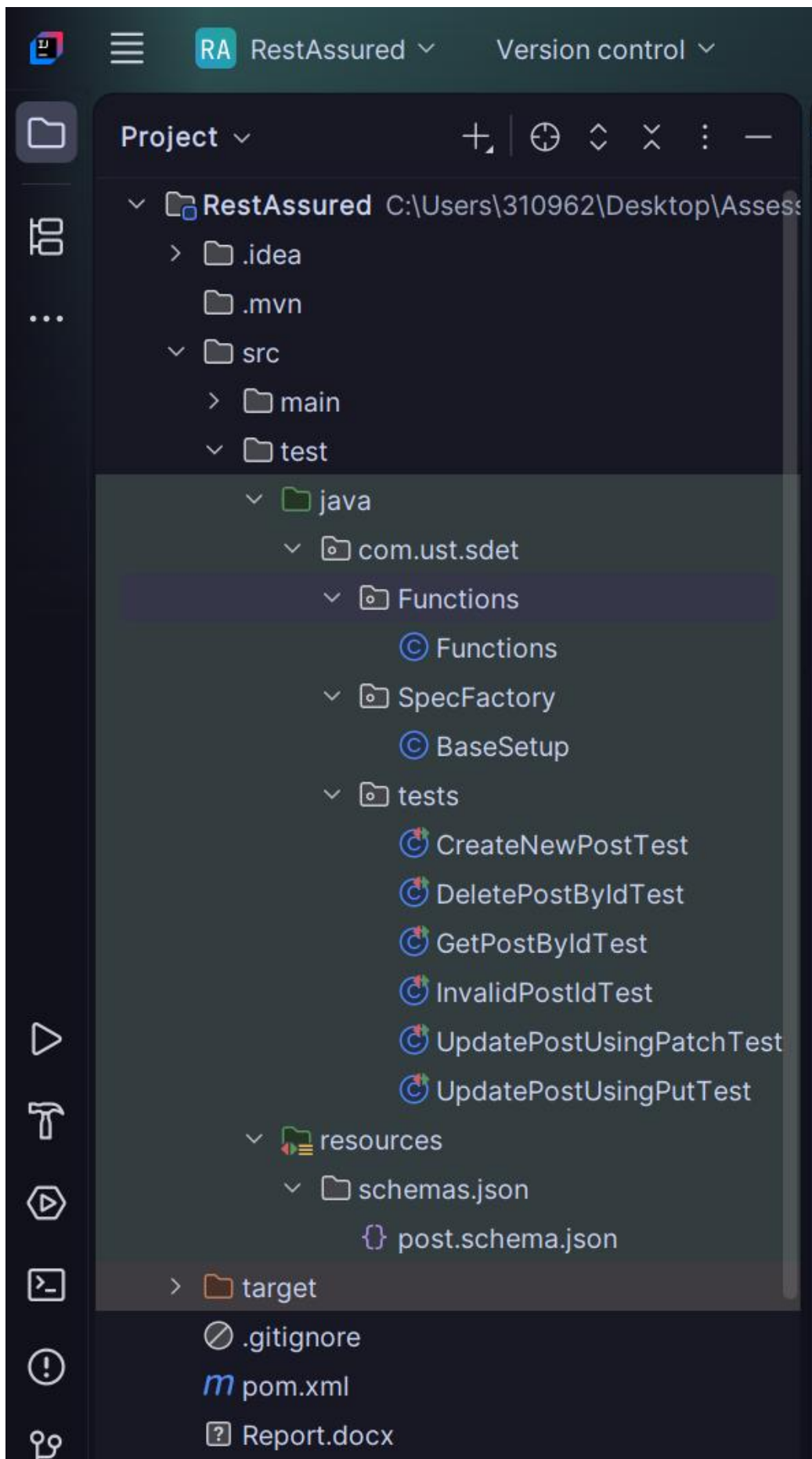
```
{  
  "$schema": "http://json-schema.org/draft-04/schema#",  
  "type": "object",  
  "required": ["id", "body", "title", "userId"],  
  "properties": {  
    "id": {  
      "type": "integer"  
    },  
    "body": {  
      "type": "string"  
    },  
    "title": {  
      "type": "string"  
    },  
    "userId": {
```

```
"type": "integer"
}
}
}
```

Environment & Setup

A new java project in IntelliJ

Folder Structure



SpecFactory - Contains the base spec builder functions

- **BaseSetup** :- Contains the base spec builder function setup (for initializing the base url) and response spec (for verifying the response code)

Functions - Contains the base spec builder functions aswell as basic post get functions

- **BaseSetup** :- Contains the base spec builder functions aswell as basic post get functions

Tests - Contains the tests being run

Pom.xml – Contains the dependencies

junit-jupiter ---> For the test Tag recognitions

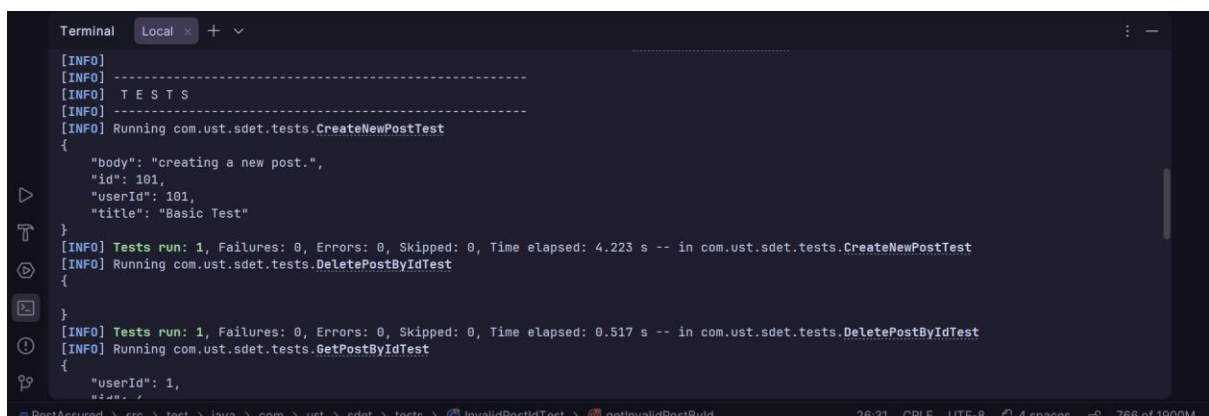
rest-assured ---> provide the functions given,when, then for the rest api calls and response handling

json-schema-validator --->provides the function for schema validation

hamcrest matchers --> provides the functions for assertions

maven-surefire plugin ---> Helps in running the Test files and setting up the configuration

Results:



```
Terminal Local x + v
[INFO] -----
[INFO] T E S T S
[INFO] -----
[INFO] Running com.ust.sdet.tests.CreateNewPostTest
{
  "body": "creating a new post.",
  "id": 101,
  "userId": 101,
  "title": "Basic Test"
}
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 4.223 s -- in com.ust.sdet.tests.CreateNewPostTest
[INFO] Running com.ust.sdet.tests.DeletePostByIdTest
{
}
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.517 s -- in com.ust.sdet.tests.DeletePostByIdTest
[INFO] Running com.ust.sdet.tests.GetPostByIdTest
{
  "userId": 1,
  "title": "Basic Test"
}
RestAssured > src > test > java > com > ust > sdet > tests > InvalidPostIdTest > getInvalidPostById 26:31 CRLF UTF-8 4 spaces 766 of 1900M
```

```
Terminal Local x
}
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.517 s -- in com.ust.sdet.tests.DeletePostByIdTest
[INFO] Running com.ust.sdet.tests.GetPostByIdTest
{
  "userId": 1,
  "id": 4,
  "title": "eum et est occaecati",
  "body": "ullam et saepe reiciendis voluptatem adipisci\nsit amet autem assumenda provident rerum culpa\nquis hic commodi nesciunt rem tenetur d
oloremque ipsam iure\nquis sunt voluptatem rerum illo velit"
}
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.231 s -- in com.ust.sdet.tests.GetPostByIdTest
[INFO] Running com.ust.sdet.tests.InvalidPostIdTest
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.216 s -- in com.ust.sdet.tests.InvalidPostIdTest
[INFO] Running com.ust.sdet.tests.UpdatePostUsingPatchTest
{
  "userId": 1,
  "id": 1,
  "title": "sunt aut facere repellat provident occaecati excepturi optio reprehenderit",
  "body": "quia et suscipit\nsuscepit recusandae consequuntur expedita et cum\nreprehenderit molestiae ut ut quas totam\nnostrum rerum est autem
sunt rem eveniet architecto"
}
PS C:\Users\310962\Desktop\Assessment\RestAssured> src > test > java > com > ust > sdet > tests > InvalidPostIdTest > getInvalidPostById
```

```
Terminal Local x + v
"userId": 1,
"id": 1,
"title": "sunt aut facere repellat provident occaecati excepturi optio reprehenderit",
"body": "quia et suscipit\nsuscepit recusandae consequuntur expedita et cum\nreprehenderit molestiae ut ut quas totam\nnostrum rerum est autem
sunt rem eveniet architecto"
}
{
  "userId": 100,
  "id": 1,
  "title": "sunt aut facere repellat provident occaecati excepturi optio reprehenderit",
  "body": "quia et suscipit\nsuscepit recusandae consequuntur expedita et cum\nreprehenderit molestiae ut ut quas totam\nnostrum rerum est autem
sunt rem eveniet architecto"
}
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.852 s -- in com.ust.sdet.tests.UpdatePostUsingPatchTest
[INFO] Running com.ust.sdet.tests.UpdatePostUsingPutTest
{
  "body": "creating a new post.",
  "id": 1,
  "userId": 1,
  "title": "Basic Test"
}
PS C:\Users\310962\Desktop\Assessment\RestAssured> src > test > java > com > ust > sdet > tests > InvalidPostIdTest > getInvalidPostById
```

```
Terminal Local x + v
[INFO] Running com.ust.sdet.tests.UpdatePostUsingPutTest
{
  "body": "creating a new post.",
  "id": 1,
  "userId": 1,
  "title": "Basic Test"
}
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 1.082 s -- in com.ust.sdet.tests.UpdatePostUsingPutTest
[INFO] Results:
[INFO] Tests run: 6, Failures: 0, Errors: 0, Skipped: 0
[INFO] BUILD SUCCESS
[INFO] Total time: 12.077 s
[INFO] Finished at: 2026-06-25T12:49:52+05:30
[INFO]
PS C:\Users\310962\Desktop\Assessment\RestAssured>
PS C:\Users\310962\Desktop\Assessment\RestAssured>
```